

LAB 02

Digital RTL Design

Combinational & Sequential Circuit Design

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MUX_3_1

1.1 RTL Design

```

1 //Behavioral Modeling
2 module mux31_behavioral(
3     input d0, d1, d2, s0 , s1,
4     output reg out
5 );
6 always @(*) begin
7     case ({s0,s1})
8         2'b00 : out = d0;
9         2'b01 : out = d1;
10        2'b10, 2'b11: out = d2;
11    endcase
12 end
13 endmodule

```

```

1 //Data Flow
2 module mux31_dataflow(
3     input d0, d1, d2, s0, s1,
4     output out
5 );
6 assign out = (s1 == 1 )? d2: (s0==0)? d0 : d2;
7
8 endmodule

```

1.2 Testbench

```

1 module mux31_tb ();
2     reg d0, d1, d2 , s0, s1;
3     wire out_dut1, out_dut2;
4
5     mux31_behavioral DUT1 (.d0(d0),.d1(d1),.d2(d2), .s0(s0), .s1(s1),.out(
6 out_dut1));
7     mux31_dataflow DUT2 (.d0(d0),.d1(d1),.d2(d2), .s0(s0), .s1(s1),.out(
8 out_dut2));
9
10    integer i;
11
12    initial begin
13        for(i=0; i<100 ; i=i+1)begin
14            d0 = $random % 2;
15            d1 = $random % 2;
16            d2 = $random % 2;
17            s0 = $random % 2;
18            s1 = $random % 2;
19            #10
20            if(out_dut1 != out_dut2)begin
21                $display("Mux 3-to-1 is not working properly!!!");

```

```

20         $stop;
21     end else begin
22         $display("Mux 3-to-1 worked Perfectly (Correct)");
23     end
24 end
25 $stop;
26 end
27
28 initial begin
29     $monitor("d0= %d, d1= %d, d2= %d, s0= %d, s1= %d, DUT1_Out= %d,
30     DUT2_Out",
31             d0, d1, d2, s0, s0 , s1);
32 end
33 endmodule

```

1.3 Simulation Waveform

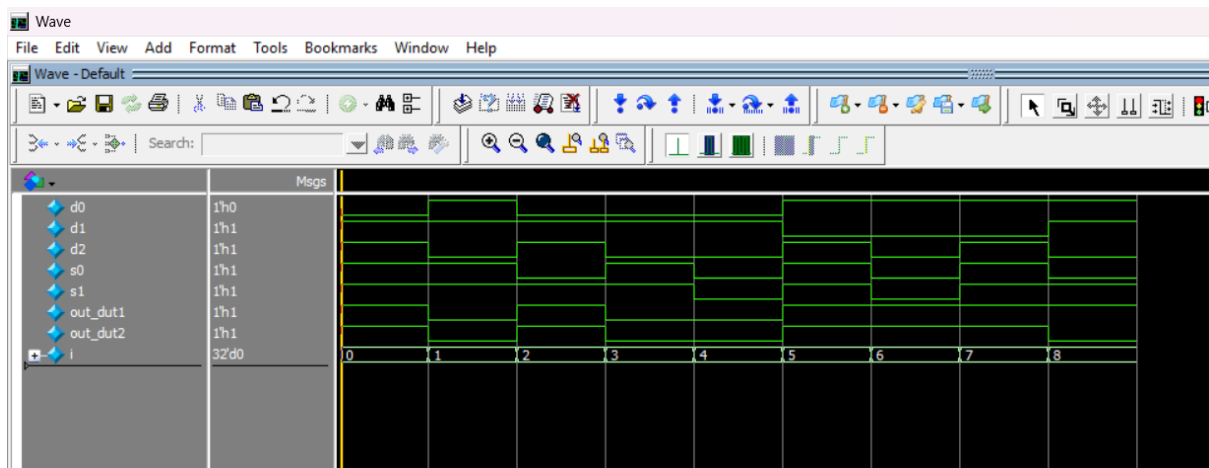


Figure 1: Simulation waveform

```

Using alternate file: ./wlfthxex93
d0= 0, d1= 1, d2= 1, s0= 1, s1= 1, DUT1_Out= 1, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 1, d1= 1, d2= 0, s0= 1, s1= 1, DUT1_Out= 1, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 0, d1= 1, d2= 1, s0= 0, s1= 0, DUT1_Out= 1, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 0, d1= 1, d2= 0, s0= 1, s1= 1, DUT1_Out= 1, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 0, d1= 1, d2= 0, s0= 0, s1= 0, DUT1_Out= 0, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 1, d1= 0, d2= 1, s0= 1, s1= 1, DUT1_Out= 1, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 1, d1= 0, d2= 0, s0= 0, s1= 0, DUT1_Out= 0, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 1, d1= 0, d2= 1, s0= 1, s1= 1, DUT1_Out= 1, DUT2_Out
Mux 3-to-1 worked Perfectly (Correct)
d0= 1, d1= 1, d2= 0, s0= 0, s1= 0, DUT1_Out= 1, DUT2_Out
Mux 3-to-1 is not working properly!!!
** Note: $stop      : mux3l_tb.v(24)
Time: 90 ns Iteration: 0 Instance: /mux3l_tb
Break in Module mux3l_tb at mux3l_tb.v line 24

```

Figure 2: Transcript

2 DFF with Clock Enable and Pre Control Signal

```
1 module dff (  
2     input E, D, clk, PRE,  
3     output reg Q  
4 );  
5  
6 always @(posedge clk or negedge PRE) begin  
7     if(!PRE)begin  
8         Q <= 1'b1;  
9     end else if (E) begin  
10        Q <= D;  
11    end  
12  
13 end  
14  
15 endmodule
```